

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCC3073 / UCCG3073 DATA SCIENCE

FRIDAY, 13 OCTOBER 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF SCIENCE (HONOURS)
STATISTICAL COMPUTING AND OPERATIONAL RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the FIRST (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

Q1. (a) Briefly discuss the business understanding stage in the data science methodology and TWO (2) reasons why it is important. (6 marks)

(b) Name and describe any TWO (2) job roles in data science. (4 marks)

(c) Consider the following linear scaled data:

-2, 2, -1.9, 1.46, 1.56, 1.02, 1.32, 0.59

(i) Determine the range of the data. (1 mark)

(ii) Given that the minimum and the maximum values of the original data are 17 and 99, inverse transform the data back to its original value. Round your answers to the nearest integer. (7 marks)

(d) For each of the following statement, identify whether it is data, information, knowledge, insight or wisdom.

(i) A flying balloon. (1 mark)

(ii) A, B, C, D (1 mark)

(iii) New Year is fall on 1st January. (1 mark)

(iv) 80% of the time in data science project was spent on cleaning the data. (1 mark)

(v) By 2030, 1 in 6 people in the world will be aged 60 years old and above. (1 mark)

(vi) Declining fertility, and increasing longevity causing elderly rise throughout the world. (1 mark)

(vii) Policies for longevity was proposed to address the whole life course and help people seize the opportunities longer lives present. (1 mark)

[Total : 25 marks]

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- Q2. (a) A company claims that their new system increases the productivity of employees by at least 20%. To test this claim, a random sample of 50 employees is selected, and their productivity levels are measured before and after using the new system. The sample average increase is found to be 18% with a standard deviation of 5%.
Conduct a hypothesis test to determine if there is sufficient evidence to support the company's claim at a 5% significance level. Determine the following.
- (i) Null and alternative hypotheses. (2 marks)
 - (ii) Standard error. Round your answer to 2 decimal points. (2 marks)
 - (iii) t test. Round your answer to 2 decimal points. (2 marks)
 - (iv) p-value. Round your answer to 4 decimal points. (2 marks)
 - (v) upper- and lower-bounds. Round your answer to 2 decimal points. (4 marks)
- (b) Using the answers of Q2(a)(ii)-(v), justify whether the null hypothesis of Q2(a)(i) can be rejected. (2 marks)
- (c) Assume that the random sample of Q2(a) is 10 employees. Calculate the t test and p-value. Round your answers to 3 decimal points. (4 marks)
- (d) Using the answers of Q2(c), justify whether the null hypothesis of Q2(a)(i) can be rejected. (2 marks)
- (e) Describe how Central Limit Theorem affects the answers of Q2(b) and Q2(d). (5 marks)
- [Total : 25 marks]

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- Q3. The following training dataset of email classified as either 'spam' or 'not spam'. A Bayesian model is used to train and test the data.

Table 3.1

Email ID	Contains "buy"	Contains "discount"	Word counting	Class
1	Yes	No	Many	Not spam
2	Yes	Yes	Many	Not spam
3	No	Yes	Little	Spam
4	Yes	Yes	Fair	Spam
5	No	No	Fair	Spam
6	No	Yes	Little	Not spam

The test data is given as follow:

Email ID	Contains "buy"	Contains "discount"	Word counting	Class
7	-	Yes	Fair	?

- State any possible errors that might occur when calculating probability for the test data. (2 marks)
 - Suggest a solution to overcome the error stated in Q3(a). (2 marks)
 - Using the solution proposed in Q3(b), calculate the probability of the Word counting attribute. Round your answer to the nearest 2 decimal points. (3 marks)
 - Using the answer of Q3(c), calculate the likelihood of Email ID 7 is not a spam email. Show the working steps in your answer. Round your answer to the nearest 2 decimal points. (6 marks)
 - Assume that the imputed value for the missing value on Email ID 7 is *no*, justify whether the likelihood of the test instance is not a spam email is higher or lower than the likelihood computed with no data imputation. (6 marks)
 - Provide TWO (2) advantages and TWO (2) disadvantages of Bayesian model. (6 marks)
- [Total : 25 marks]

SECTION B (Choose only ONE question)

- Q4. A Naïve Bayes classification model was developed to detect credit card fraud of their customers. Table 4.1 shows the prediction made by the model on the test data. The value 'Yes' indicates a fraudulent transaction made on the credit card and the value 'No' indicates a normal transaction.

Table 4.1

ID	Target	Prediction
1	No	No
2	No	No
3	No	Yes
4	No	Yes
5	No	No
6	No	No
7	No	Yes
8	No	No
9	No	No
10	Yes	No

ID	Target	Prediction
11	Yes	No
12	Yes	No
13	Yes	No
14	Yes	No
15	Yes	Yes
16	Yes	No
17	Yes	Yes
18	Yes	No
19	Yes	Yes
20	Yes	Yes

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate true positives (TPs), true negatives (TNs), false positive (FPs) and false negatives (FNs). (4 marks)
- (c) Using the computed results in Q4(b), calculate the following evaluation measures. Round your answer to the nearest 2 decimal points.
- (i) Classification accuracy rate. (1 mark)
 - (ii) Precision. (1 mark)
 - (iii) Recall. (1 mark)
 - (iv) Specificity. (1 mark)
 - (v) F_1 score. (2 marks)
- (d) Using the computed results in Q4(a) and Q4(c), justify whether you would propose the model to the company. (7 marks)
- (e) Suggest TWO (2) ways to overcome issues you have observed in Q4(d). (6 marks)

[Total : 25 marks]

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- Q5. Two (2) decision tree models were used to predict cancer diagnostic on two cohort of patients. Table 5.1 shows the prediction results of the models on the test data. Category healthy indicates the healthy patients and category cancer indicates cancer patients.

Table 5.1

N = 100		Predicted values			
		Model 1		Model 2	
		Healthy	Cancer	Healthy	Cancer
Target/ Actual	Healthy	47	0	24	23
	Cancer	24	29	9	44

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 2 decimal points.
- (i) Misclassification rate. (3 marks)
- (ii) Recall. (3 marks)
- (iii) Specificity. (3 marks)
- (iv) Precision. (3 marks)
- (c) Using the computed results in Q5(a) and Q5(b), justify whether you would propose Model 1 to the hospital. (3 marks)
- (d) Model 1 uses equal cost and Model 2 uses unequal cost where the cost for false positive ($Cost_{FP}$) is 5 and the cost for true positive ($Cost_{TP}$) is -15.
- (i) Determine the model profit per record for both models. (3 marks)
- (ii) Justify whether the unequal cost model is better than the baseline performance and the equal cost model. (5 marks)
- [Total : 25 marks]

APPENDIX

Proximity measures	
Euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ Manhattan distance $d(x, y) = \sum_{i=1}^n x_i - y_i $ Supremum distance $d(x, y) = \max(x_i - y_i)$	Hamming distance $d(a, b) = (a \oplus b)$ Simple matching coefficient $SMC = \frac{\text{num of matching attribute values}}{\text{num of attributes}}$ Jaccard coefficient $J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$ Cosine similarity $\cos(x, y) = \frac{x \cdot y}{ x y }$
Regression	
Straight line $\hat{y} = \beta_0 + \beta_1 x$ Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ Residuals $\hat{\epsilon} = y - \hat{y}$ $s^2 = \frac{\sum \hat{\epsilon}^2}{p - 2}$ $s = \sqrt{s^2}$ t value $t = \frac{\hat{\beta}}{SE}$	Coefficient of Determination $R^2 = \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{(p \sum x^2 - (\sum x)^2) * (p \sum y^2 - (\sum y)^2)}} \right]^2$ Pearson correlation $R = \sqrt{R^2}$ Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$ $SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Naïve Bayes	
Bayes' rule $P(H E) = \frac{P(E H)P(H)}{P(E)}$ Probability density function $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$	Laplace correction $\frac{x_i + 1}{c + \alpha}$

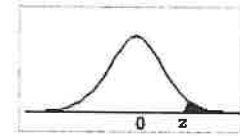
Association Rules	
Association rule $X \rightarrow Y$ Support $s = \frac{freq(X,Y)}{N}$	Confidence $c = \frac{freq(X,Y)}{freq(X)}$ Lift $l = \frac{Confidence(X,Y)}{Support(X)}$

Evaluation metrics	
Classification accuracy $acc = \frac{TN + TP}{TN + TP + FN + FP}$ Misclassification rate $1 - acc$ Sensitivity / Recall $recall = \frac{TP}{TP + FN}$ Specificity $spec = \frac{TN}{TN + FP}$ False discovery rate $FDR = \frac{FP}{FP + TP}$	Precision / Positive Predictive Value $PPV = \frac{TP}{TP + FP} = 1 - FDR$ F1 score $F_1 = \frac{2TP}{2TP + FP + FN}$ Overall model cost $TN.Cost_{TN} + FP.Cost_{FP} + FN.Cost_{FN} + TP.Cost_{TP}$ Model cost per case $\frac{Overall\ model\ cost}{N}$ Model profit per case $-model\ cost\ per\ case$

Miscellaneous	
Linear scaling for [a, b] $x' = (b - a) \frac{x - \min(x)}{\max(x) - \min(x)} + a$ z-score normalisation $x' = \frac{x - \mu}{\sigma}$ Mean, $\bar{x}$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ Z-test $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$ t-test $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$	Inverse linear scaling for [a, b] $x = (x' - a) * \frac{\max(x) - \min(x)}{b - a} + \min(x)$ Population standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ Sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$ Standard error $SE = \frac{\sigma}{\sqrt{n}}$ Confidence intervals $CI = \bar{x} \pm z * SE$

UCCC3073 / UCCG3073 DATA SCIENCE**Normal Curve Areas (Standard normal probability in right-hand tail)**for negative values of z , areas are found by symmetry confidence level to Z-score

90%	1.645
95%	1.96
98%	2.33
99%	2.575



z	Second decimal place of z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

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t Distribution: Critical Values of t								
Degree of Freedom	1-tailed α	Significant level						
		0.25	0.2	0.15	0.1	0.05	0.025	0.01
2-tailed α		0.5	0.4	0.3	0.2	0.1	0.05	0.02
1		1.000	1.376	1.963	3.078	6.314	12.706	31.821
2		0.816	1.061	1.386	1.886	2.920	4.303	6.965
3		0.765	0.978	1.250	1.638	2.353	3.182	4.541
4		0.741	0.941	1.190	1.533	2.132	2.776	3.747
5		0.727	0.920	1.156	1.476	2.015	2.571	3.365
6		0.718	0.906	1.134	1.440	1.943	2.447	3.143
7		0.711	0.896	1.119	1.415	1.895	2.365	2.998
8		0.706	0.889	1.108	1.397	1.860	2.306	2.896
9		0.703	0.883	1.100	1.383	1.833	2.262	2.821
10		0.700	0.879	1.093	1.372	1.812	2.228	2.764
11		0.697	0.876	1.088	1.363	1.796	2.201	2.718
12		0.695	0.873	1.083	1.356	1.782	2.179	2.681
13		0.694	0.870	1.079	1.350	1.771	2.160	2.650
14		0.692	0.868	1.076	1.345	1.761	2.145	2.624
15		0.691	0.866	1.074	1.341	1.753	2.131	2.602
16		0.690	0.865	1.071	1.337	1.746	2.120	2.583
17		0.689	0.863	1.069	1.333	1.740	2.110	2.567
18		0.688	0.862	1.067	1.330	1.734	2.101	2.552
19		0.688	0.861	1.066	1.328	1.729	2.093	2.539
20		0.687	0.860	1.064	1.325	1.725	2.086	2.528
21		0.686	0.859	1.063	1.323	1.721	2.080	2.518
22		0.686	0.858	1.061	1.321	1.717	2.074	2.508
23		0.685	0.858	1.060	1.319	1.714	2.069	2.500
24		0.685	0.857	1.059	1.318	1.711	2.064	2.492
25		0.684	0.856	1.058	1.316	1.708	2.060	2.485
26		0.684	0.856	1.058	1.315	1.706	2.056	2.479
27		0.684	0.855	1.057	1.314	1.703	2.052	2.473
28		0.683	0.855	1.056	1.313	1.701	2.048	2.467
29		0.683	0.854	1.055	1.311	1.699	2.045	2.462
30		0.683	0.854	1.055	1.310	1.697	2.042	2.457
31		0.682	0.853	1.054	1.309	1.696	2.040	2.453
32		0.682	0.853	1.054	1.309	1.694	2.037	2.449
33		0.682	0.853	1.053	1.308	1.692	2.035	2.445
34		0.682	0.852	1.052	1.307	1.691	2.032	2.441
35		0.682	0.852	1.052	1.306	1.690	2.030	2.438
36		0.681	0.852	1.052	1.306	1.688	2.028	2.434
37		0.681	0.851	1.051	1.305	1.687	2.026	2.431
38		0.681	0.851	1.051	1.304	1.686	2.024	2.429
39		0.681	0.851	1.050	1.304	1.685	2.023	2.426
40		0.681	0.851	1.050	1.303	1.684	2.021	2.423
41		0.681	0.850	1.050	1.303	1.683	2.020	2.421
42		0.680	0.850	1.049	1.302	1.682	2.018	2.418
43		0.680	0.850	1.049	1.302	1.681	2.017	2.416
44		0.680	0.850	1.049	1.301	1.680	2.015	2.414
45		0.680	0.850	1.049	1.301	1.679	2.014	2.412
46		0.680	0.850	1.048	1.300	1.679	2.013	2.410
47		0.680	0.849	1.048	1.300	1.678	2.012	2.408
48		0.680	0.849	1.048	1.299	1.677	2.011	2.407
49		0.680	0.849	1.048	1.299	1.677	2.010	2.405
50		0.679	0.849	1.047	1.299	1.676	2.009	2.403
60		0.679	0.848	1.045	1.296	1.671	2.000	2.390
70		0.678	0.847	1.044	1.294	1.667	1.994	2.381
80		0.678	0.846	1.043	1.292	1.664	1.990	2.374
90		0.677	0.846	1.042	1.291	1.662	1.987	2.368
100		0.677	0.845	1.042	1.290	1.660	1.984	2.364
150		0.676	0.844	1.040	1.287	1.655	1.976	2.351
200		0.676	0.843	1.039	1.286	1.653	1.972	2.345
∞		0.675	0.842	1.036	1.282	1.645	1.960	2.327